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Assignment number and name: Task 3 – Launch a Free-Tier EC2 Instance, Connect via SSH, Install Apache, and Serve a Custom Webpage

Submission date: 17 September 2026

AWS Region Used: us-east-1(N. Virginia)

Objective

To launch a Free Tier eligible Amazon Linux 2023 EC2 instance, configure secure SSH and HTTP access, connect to the instance using SSH, install and configure Apache HTTP Server, serve a custom webpage, verify the webpage through the public IP address, test the public IP change after stopping and restarting the instance, and finally terminate the instance after completion.

Go to EC2 → Launch Instance.

Name: web-server-Varun

AMI: Amazon Linux 2023 (free tier eligible).

Type: t2.micro.

Create a new key pair (RSA, .pem). Download it.

Security Group:

Allow SSH port 22 from My IP.

Allow HTTP port 80 from Anywhere.

Launch instance.

Once running, connect via SSH:

```
chmod 400 your-key.pem
ssh -i "your-key.pem" ec2-user@<public-ip>
```

Run the following commands:

```
sudo yum update -y
```

```
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
echo "<h1>Hello from EC2 - [YourName]</h1>" | sudo
tee /var/www/html/index.html
```

Open a browser. Visit:

```
http://<public-ip>
```

The custom webpage should load.

Go back to EC2 console.

Stop the instance. Note that the public IP is now empty.

Start it again. A new public IP is assigned.

Terminate the instance after completing and screenshotting all steps.

Deliverables

Screenshots of:

- Running EC2 instance
- Security group rules
- SSH terminal session with executed commands
- Apache installed and running
- Custom webpage in browser
- Instance stopped (public IP removed)
- Instance restarted (new public IP)
- Instance terminated

Step 1 – Launched EC2 Instance

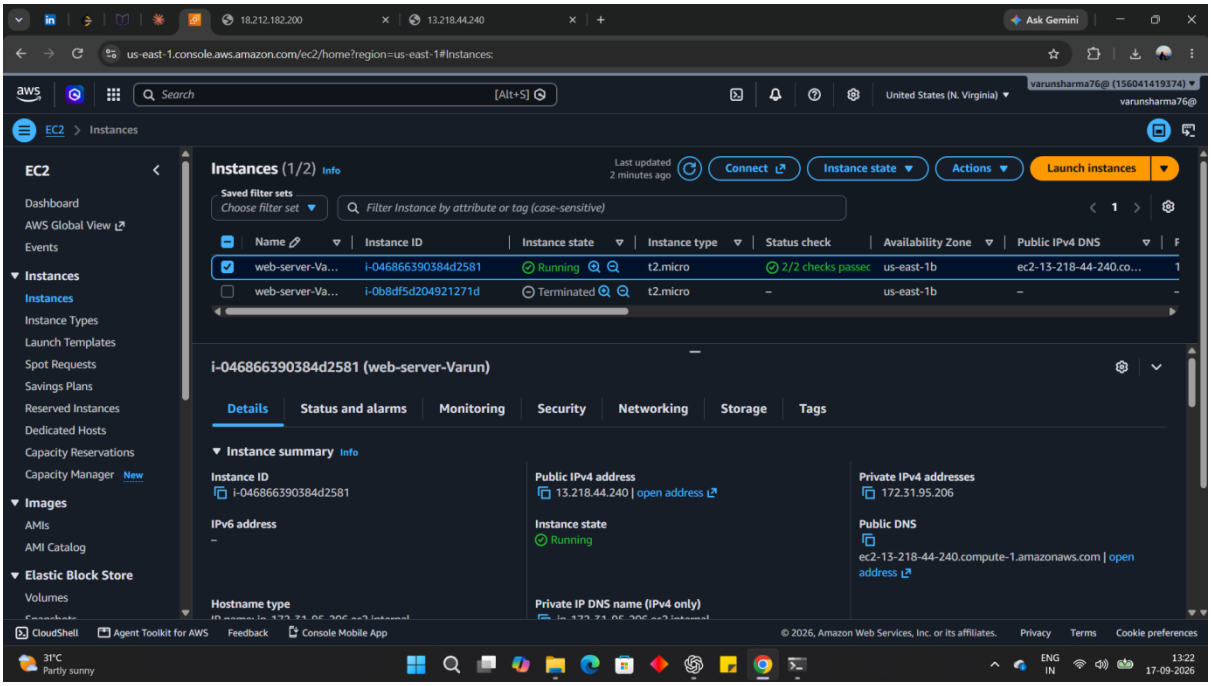
Action

Launched an EC2 instance named: web-server-Varun

The instance was configured with Amazon Linux 2023 and the t2.micro instance type.

A new RSA key pair named: **web-server-key**

was created in .pem format for SSH access.



Step 2 – Configured Security Group

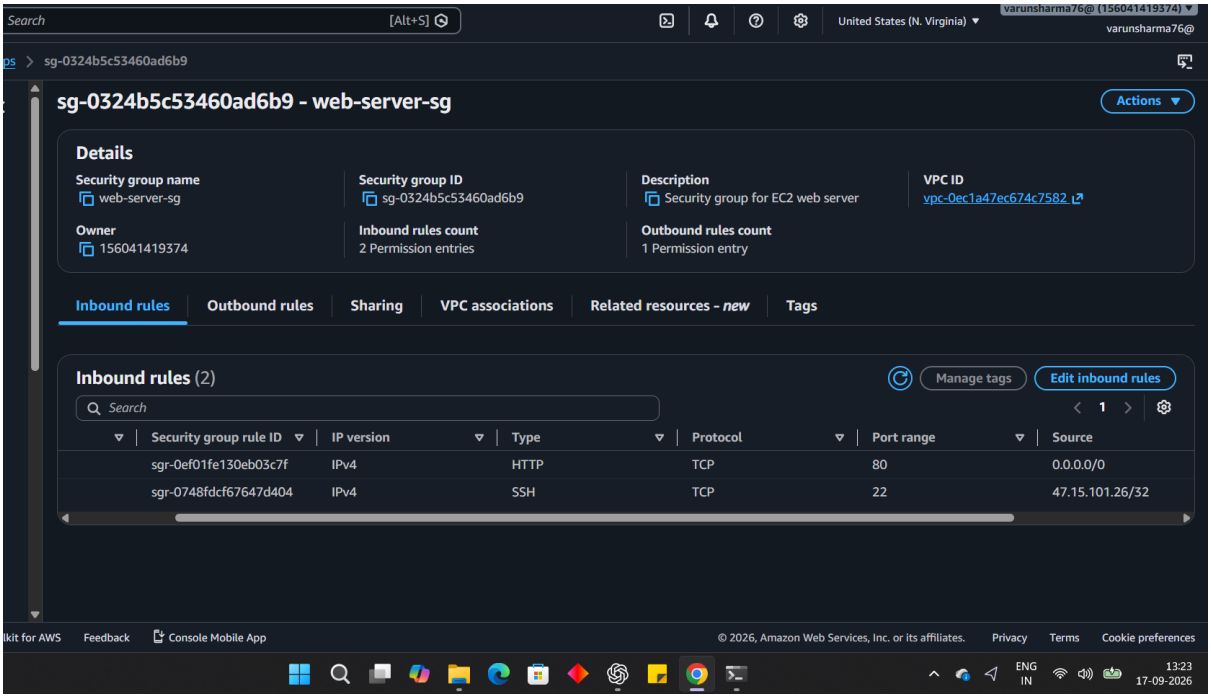
Action

Created a security group named: **web-server-sg**

Configured the following inbound rules:

Type	Protocol	Port	Source
SSH	TCP	22	My IP
HTTP	TCP	80	Anywhere IPv4

Confirmation



Step 3 – Connected to EC2 Using SSH

Action

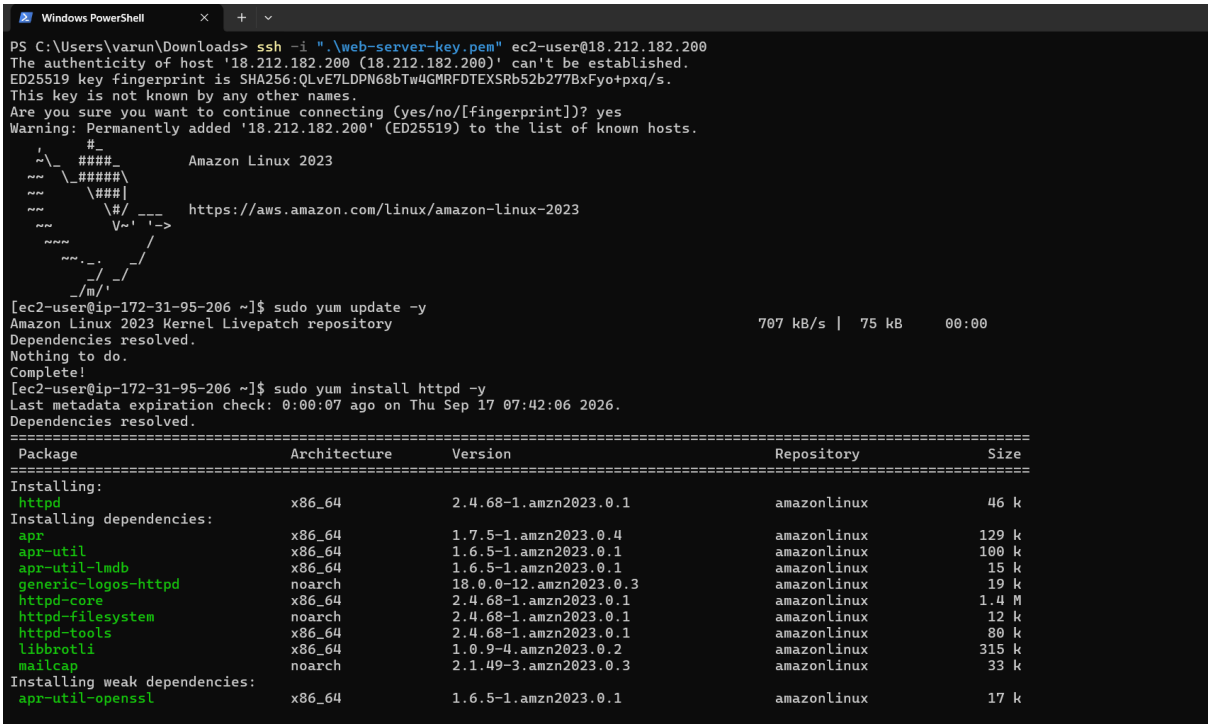
Connected to the Amazon Linux 2023 EC2 instance using the ec2-user account

and the downloaded private key.

Command executed:

ssh -i ".\web-server-key.pem" ec2-user@18.212.182.200

Confirmation



Step 4 – Installed and Configured Apache

Action:

Updated the system packages, installed Apache HTTP Server, started the Apache service, and enabled it to start automatically after system reboot.

Commands:

sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd

Confirmation

```
Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.68-1.amzn2023.0.1.x86_64
  httpd-core-2.4.68-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64                 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                     mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-95-206 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-95-206 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-95-206 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-09-17 07:42:30 UTC; 15s ago
       Docs: man:httpd.service(8)
   Main PID: 27122 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1113)
     Memory: 13.3M
        CPU: 55ms
     CGroup: /system.slice/httpd.service
             ├─27122 /usr/sbin/httpd -DFOREGROUND
             ├─27123 /usr/sbin/httpd -DFOREGROUND
             ├─27124 /usr/sbin/httpd -DFOREGROUND
             ├─27143 /usr/sbin/httpd -DFOREGROUND
             └─27144 /usr/sbin/httpd -DFOREGROUND

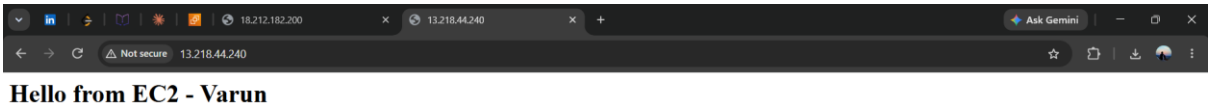
Sep 17 07:42:30 ip-172-31-95-206.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 17 07:42:30 ip-172-31-95-206.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 17 07:42:30 ip-172-31-95-206.ec2.internal httpd[27122]: Server configured, listening on: port 80
[ec2-user@ip-172-31-95-206 ~]$ echo "<h1>Hello from EC2 - Varun</h1>" | sudo tee /var/www/html/index.html
<h1>Hello from EC2 - Varun</h1>
[ec2-user@ip-172-31-95-206 ~]$ cat /var/www/html/index.html
<h1>Hello from EC2 - Varun</h1>
[ec2-user@ip-172-31-95-206 ~]$
Broadcast message from root@localhost (Thu 2026-09-17 07:47:14 UTC):
```

Step 5 – Served Custom Webpage

Action

Created a custom HTML webpage and accessed it through the EC2 public IPv4 address using a web browser.

Confirmation



Hello from EC2 - Varun